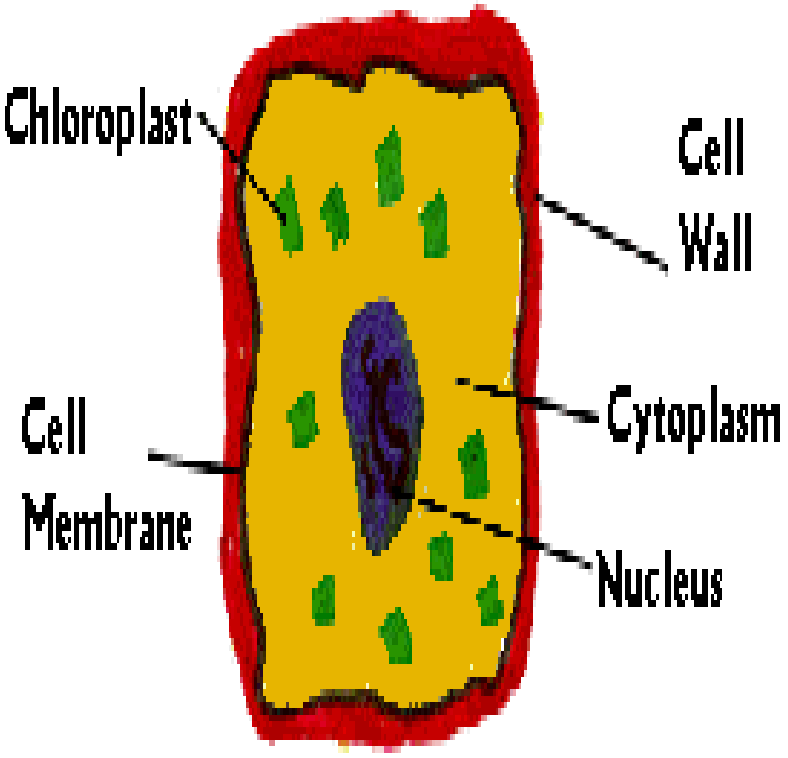Plant Cell